Type 3: Software Project

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Zero Trust–Enabled SDN for Secure Load Balancing in Edge Networks

23CSE498 – Project Phase 2 (Panel Review 1) | Team Number: 50

S.No	Register Number	Student Name
1	CB.SC.U4CSE23262	K R Jeevan Raj
2	CB.SC.U4CSE23303	ARAVIND KUMAR M
3	CB.SC.U4CSE23307	Arjun Krishnakumar
4	CB.SC.U4CSE23308	Ashwin Kumar RM
5	CB.SC.U4CSE23544	Bhuvanesh S

Guide: **Dr.Radhika N**, Professor, Department of CSE

September 7, 2026

1. Software Architecture & Module Design

Rubric: UML Diagrams, DB Schema, APIs, Module Interaction & Deployment (5 Marks)

Modular Decomposition & Contracts:

- **Module Breakdown:** Auth, Data Layer, Business Engine, API Gateway.
- **Core API Specifications:**
  - `POST /api/v1/auth/login`
  - `GET /api/v1/resource/query`
- **Database Schema:** Entity models, relations, indices.
- **Deployment Architecture:** Containers (Docker), CI/CD pipeline, environment config.

[Insert Software Architecture / UML Diagram]

(Show: Frontend ↔ Backend Services ↔ Database/Cache)

2. Implementation Progress ($\approx$60%) & Module Functionality

Rubric: Backend, Frontend, Database & Core Modules Functional (8 Marks)

- **Software Implementation Status ($\approx$60% Completed):**

Subsystem / Layer	Current Implementation State	Status
Frontend UI	Responsive views, API client, router setup	Functional
Backend Server & APIs	Core controllers, middleware, CRUD logic	Functional
Database Layer	Schema migrations, ORM entities, queries	Integrated
System Integration	Frontend-to-backend API connectivity	Integrated

- **System Demonstration Evidence:** Working feature walkthrough, API test results (Postman/cURL), GitHub repository commits.

3. Technical Knowledge & Engineering Decisions

Rubric: Software Architecture, Implementation, Integration & Tech Stack (5 Marks)

- **Architectural Patterns:** Clean Architecture / Layered / Microservices pattern justification.
- **Design Patterns & Code Structure:** Factory, Singleton, Dependency Injection, Repository pattern.
- **Data Integrity & Concurrency:** ACID transactions, thread-safety, connection pooling, indexing.
- **Error Handling & Logging:** Structured logging, exception hierarchies, graceful failure handling.
- **Preliminary Testing:** Unit testing baseline, integration tests, and performance benchmarks.

4. Technical Enrichment Activity Choices

Professional Online Certification Details (Individual Mandatory: 10–20 Hours)

- **Individual MOOC / Online Certification Selection:**

S.No	Student Name	Chosen MOOC / Course Title	Platform	Hours
1	K R Jeevan Raj	[Course Title relevant to Project]	Coursera	15 Hrs
2	ARAVIND KUMAR M	[Course Title relevant to Project]	NPTEL	20 Hrs
3	Arjun Krishnakumar	[Course Title relevant to Project]	edX / Coursera	12 Hrs
4	Ashwin Kumar RM	[Course Title relevant to Project]	AWS / Google	16 Hrs
5	Bhuvanesh S	[Course Title relevant to Project]	AWS / Google	16 Hrs

- **Requirement:** Each student must complete one approved 10–20 hour online certification from a recognized platform (Coursera, NPTEL, edX, etc.) relevant to the project.

References I

Thank You